

.NET Lab: NET High-CPU Incident Analysis

Duration : 30 Min.

Scenario

A production service is consuming 100% CPU on all cores, causing latency SLAs breaches. You need to investigate the hotspot.

Task Objectives

Perform the following actions inside the `student_workspace` directory: - Use `dotnet-trace` to capture thread execution traces, isolate high-utilization hot paths and methods, and document the code remediation.

Instructions to Perform the Task

1. When your workspace loads in **VS Code**, use the **Explorer** panel on the left to locate your files.
2. Navigate to and click the specific files mentioned in the Task Objectives.
3. Make the necessary code edits in the editor.
4. Press `Ctrl + S` (Windows) or `Cmd + S` (Mac) to save your changes.
5. If you need to run commands, open the built-in terminal by clicking **Terminal > New Terminal** from the top menu.

Validation

Once you have saved your files and are ready to submit, return to the platform dashboard and click the **"Submit Final"** button. This task is evaluated manually by the grading team.

Evaluation Rubric:

1. `dotnet-trace` collection file capture execution: 5 marks
2. CPU hotspots call tree analysis and method isolation: 5 marks

Total Score: 10 Marks

Important Notes

- This is a manually evaluated question.
- Ensure all logs and configurations are fully saved in your workspace folder before submitting.